

CSE B.Tech Planner Website

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Introduction

This project is a web-based platform that helps learning Computer Science students to

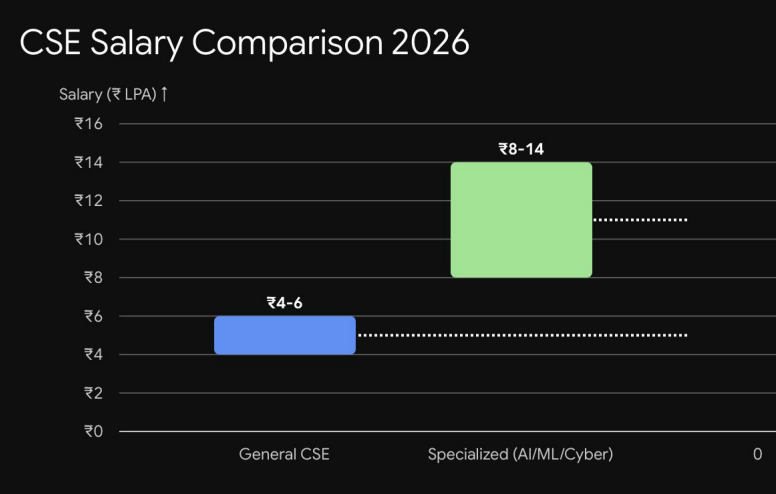
1. Explore all available popular specializations
2. Take a skill-assessment survey
3. Receive a personalized roadmap
 - a. With concise structure
 - b. With industry-required resources
 - c. With interim survey to evolve to the user's needs

Some of the various popular B Tech Computer Science Specializations are:

1. AI/ML
2. Data Science
3. Cybersecurity
4. Full Stack Web Development
5. Cloud Computing
6. Cybersecurity
7. IoT
8. Blockchain
9. DevOps
10. UI/UX Design
11. Mobile App Development
12. Computer Networks

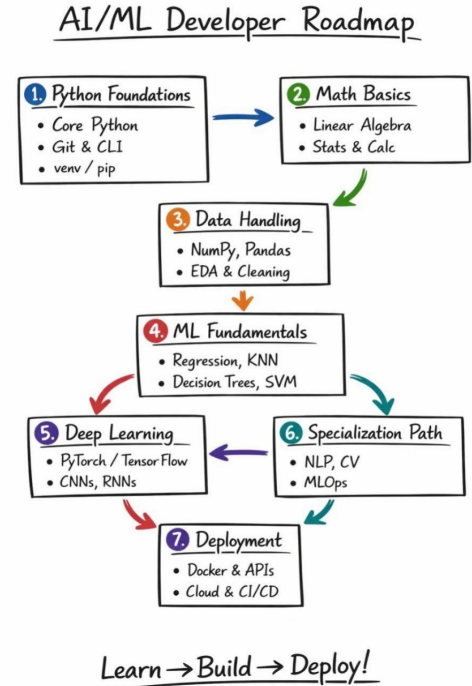
Motivation

- Students frequently lack exposure to the full breadth of CSE specializations (e.g., AI/ML, Cloud, Cybersecurity, DevOps) early in their studies, discovering their true interests too late
- In 2026, general CSE graduates starting in service companies earn ₹4–6 LPA, whereas specialized graduates in AI/ML, Cybersecurity, or Cloud secure starting packages of ₹8–14 LPA
- Industry data indicates that while 71.5% of engineering graduates are technically "employable", most lack the verified, hands-on domain competencies demanded by modern technical recruiters



Existing Systems

1. Static Roadmaps
 - a. Limitations :
 - i. Non-personalized (one size fits all)
 - ii. Entirely static (not helpful when user want to switch interests)
 - iii. No competency assessment tied to a specialization
 - iv. Usually outdated
2. General Learning Platforms - Coursera
 - a. Limitations :
 - i. Only a course catalog, not a guidance system
 - ii. No competency assessment tied to a specialization (sometimes users take courses that was too challenging)
 - iii. engagement-driven , not academically structured
 - iv. No unified tracking system
 - v. Paid model

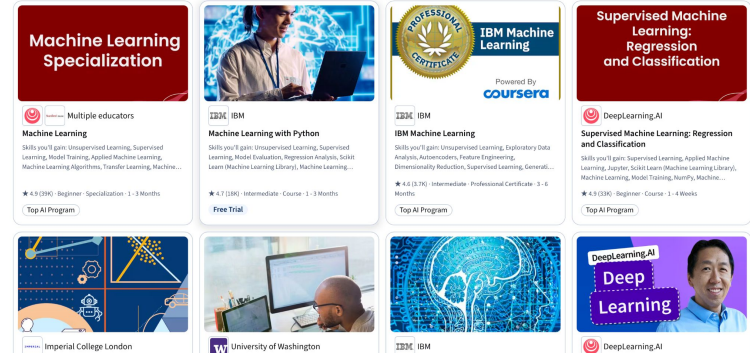


Existing Systems

3. Placement Preparation Tools :

a. Limitations :

- i. Highly focused on final-year interviews
- ii. Focuses on student's overall capability, not exclusively technical knowledge - communication skills, coding skills, teamwork, etc.
- iii. Neglects year-by-year development for the student

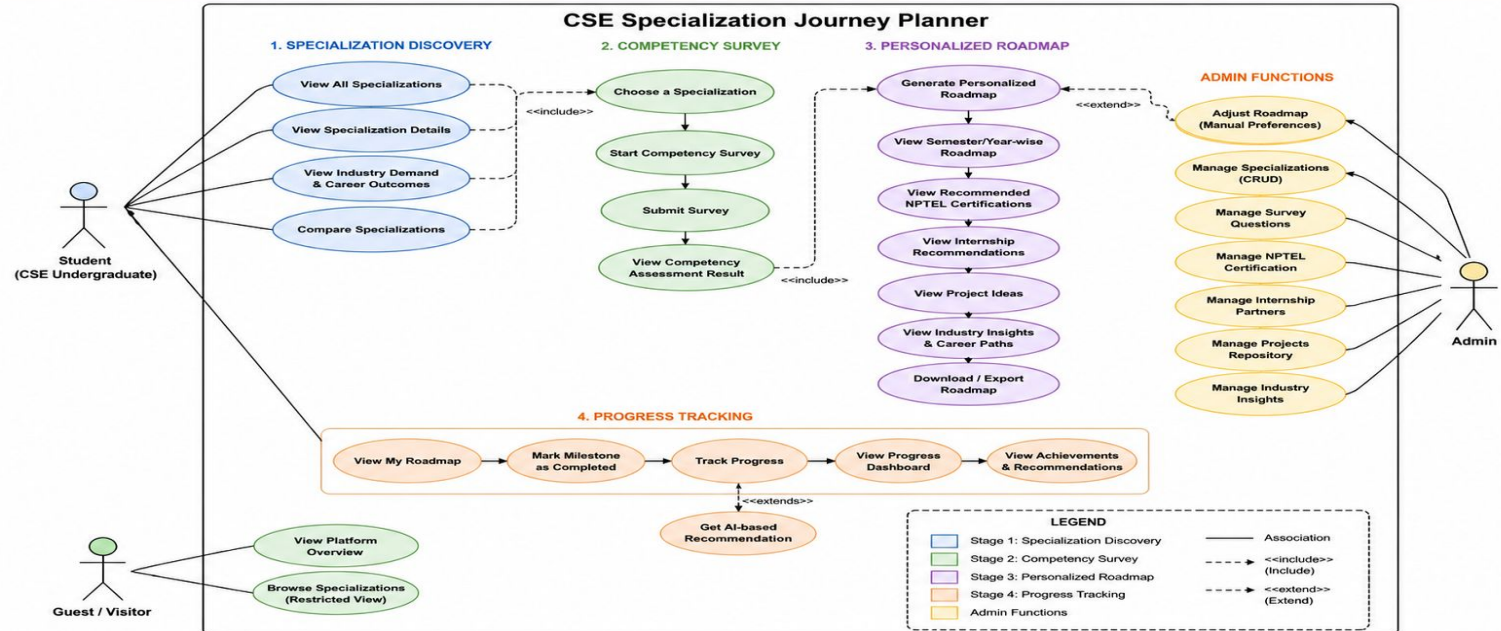


Methodology

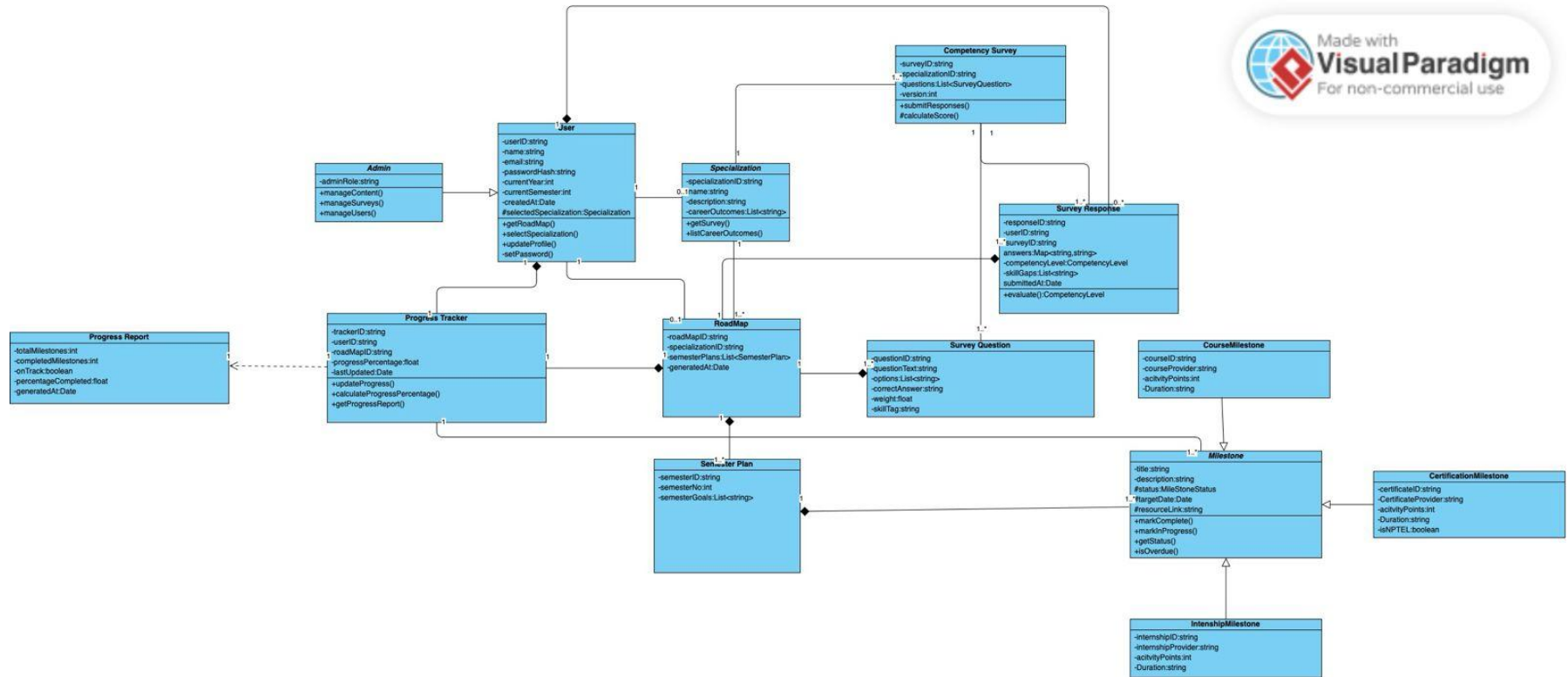
1. Data curation and storage : Data will be stored in MongoDB (Document based storage). Data will include set of courses, internships, and all opportunities for each Computer Science specialization
2. Competency assessment :
 - a. Dynamic competency survey for each specialization
 - b. Rule-based weighted scoring algorithm (due to feasibility)
3. Roadmap Generation:
 - a. Pre-curated admin-maintained repository of respective NPTEL courses, certifications, other MOOC courses, internship opportunities, etc.
4. Progress Tracking:
 - a. Authentication and authorization for users
 - b. Each user has a unique dashboard and progress tracking
5. **Tech Stack**
 - a. Java Spring Boot (backend) + MongoDB(database) + React.js (frontend)
 - i. Spring Boot - RESTFUL API layer
 - ii. [React.js](#) - ability to reuse UI state elements, popularity, and prior knowledge
 - iii. MongoDB - document-based model, provide nested hierarchical structure

Use Case Diagram

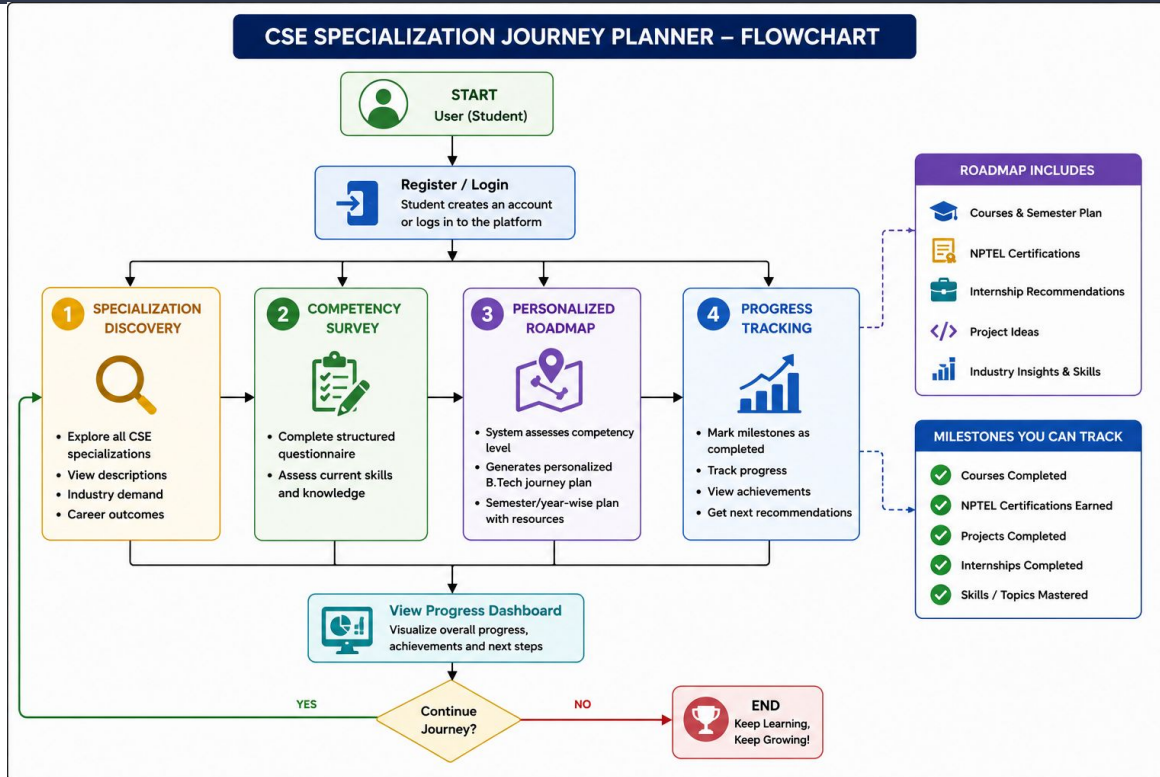
Use Case Diagram



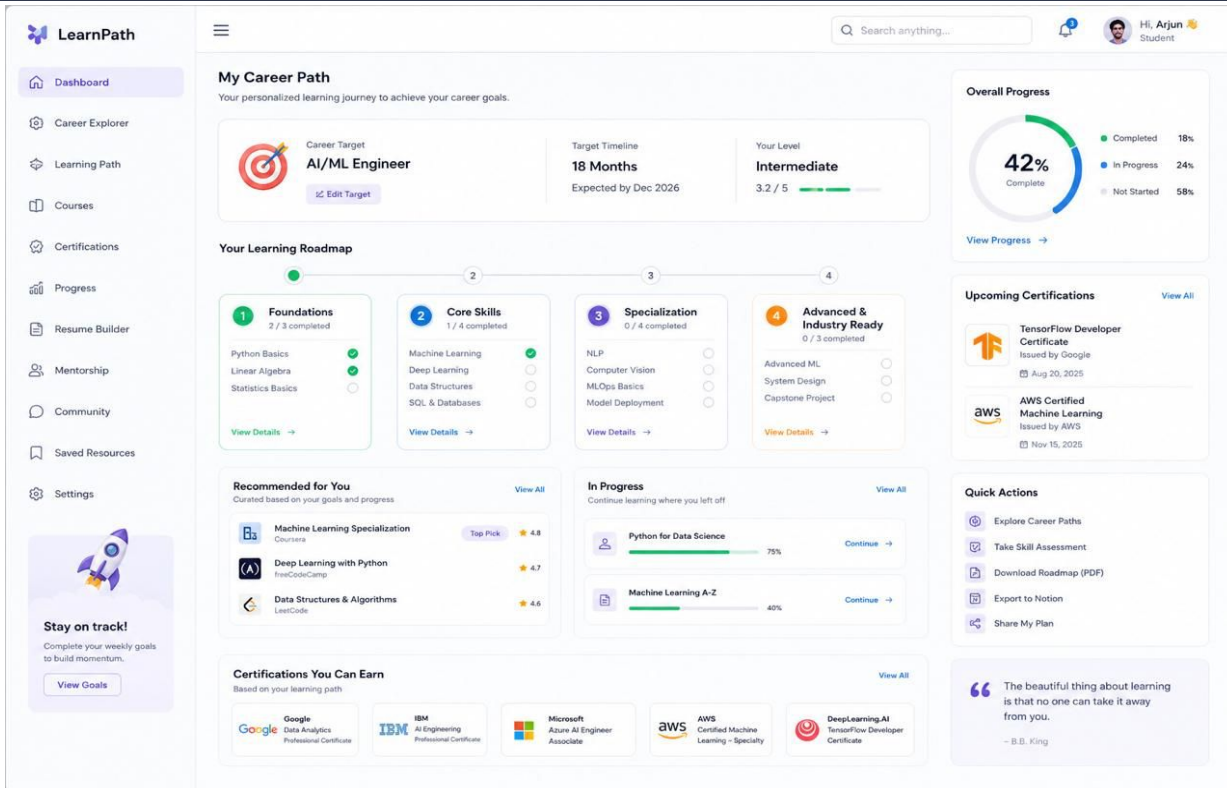
Class Diagram



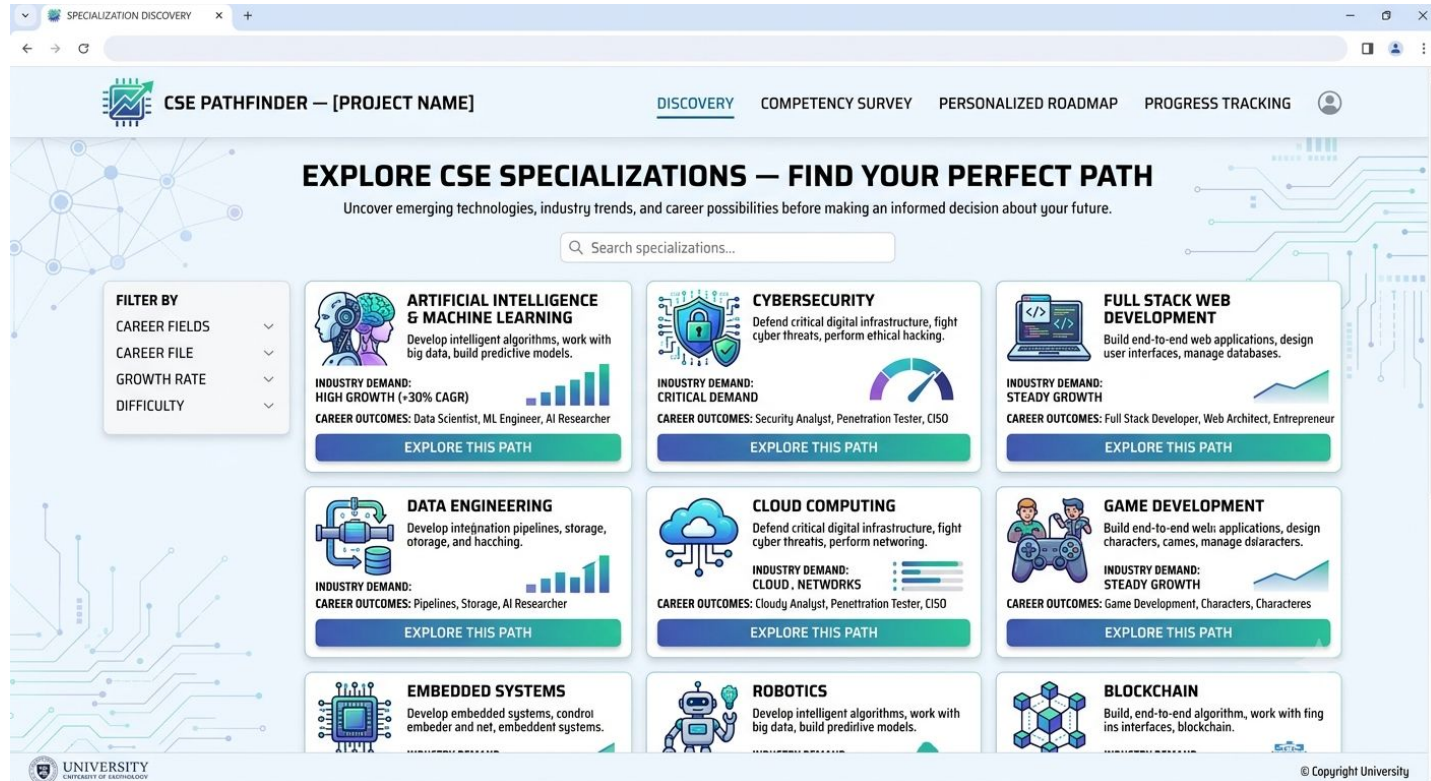
Flow Chart



GUI Design - Page 1



GUI Design – Page 2



GUI Design – Page 3

COMPETENCY SURVEY

CSE PATHFINDER — [PROJECT NAME]

DISCOVERY COMPETENCY SURVEY PERSONALIZED ROADMAP PROGRESS TRACKING

COMPETENCY ASSESSMENT: ESTABLISHING YOUR BASELINE

Self-evaluate your foundational CSE skills to tailor your personalized journey. Please answer all questions honestly.

15%

ASSESSMENT SECTIONS

Programming Fundamentals	1/5
Data Structures	1/4
Algorithms	0/3
Mathematics	0/3

SECTION 1: PROGRAMMING FUNDAMENTALS

Q1. What will be the output of the following Python code snippet?

```
x = [1, 2, 3]
x.append([4, 5])
print(len(x))
```

☐ A. 3
☒ B. 4
☐ C. 5
☐ D. Error

Q2. Implement a function to find the maximum value in an integer array.

```
1 def find_max(arr):
2     # Your code here
3     pass
4
```

RUN CODE SUBMIT SOLUTION

SECTION 2: DATA STRUCTURES

Q3. Which of the following are typically implemented as non-linear data structures? Select all that apply.

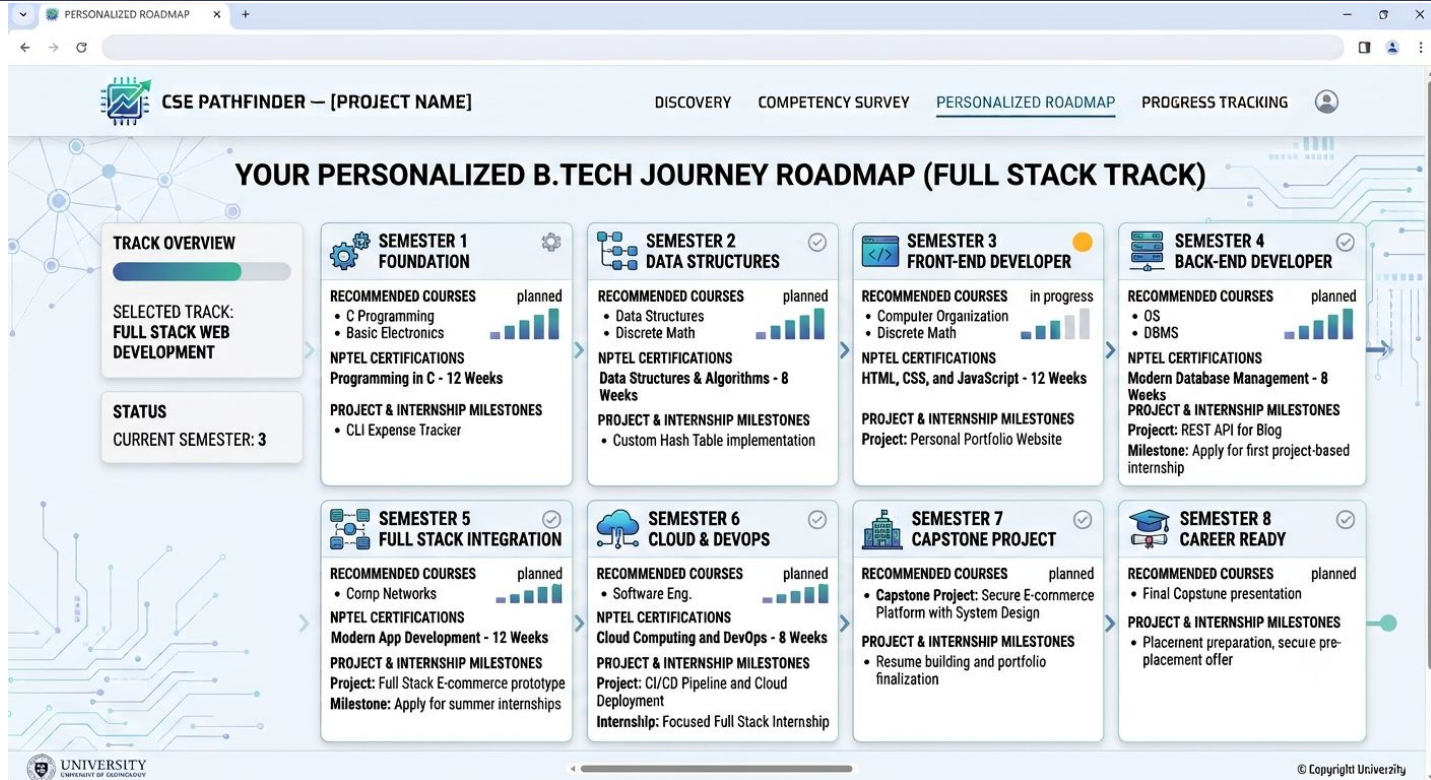
☐ Stack ☐ Queue ☐ Hash Table ☒ Graph ☐ Linked List ☒ Binary Tree

SAVE & CONTINUE SUBMIT ASSESSMENT

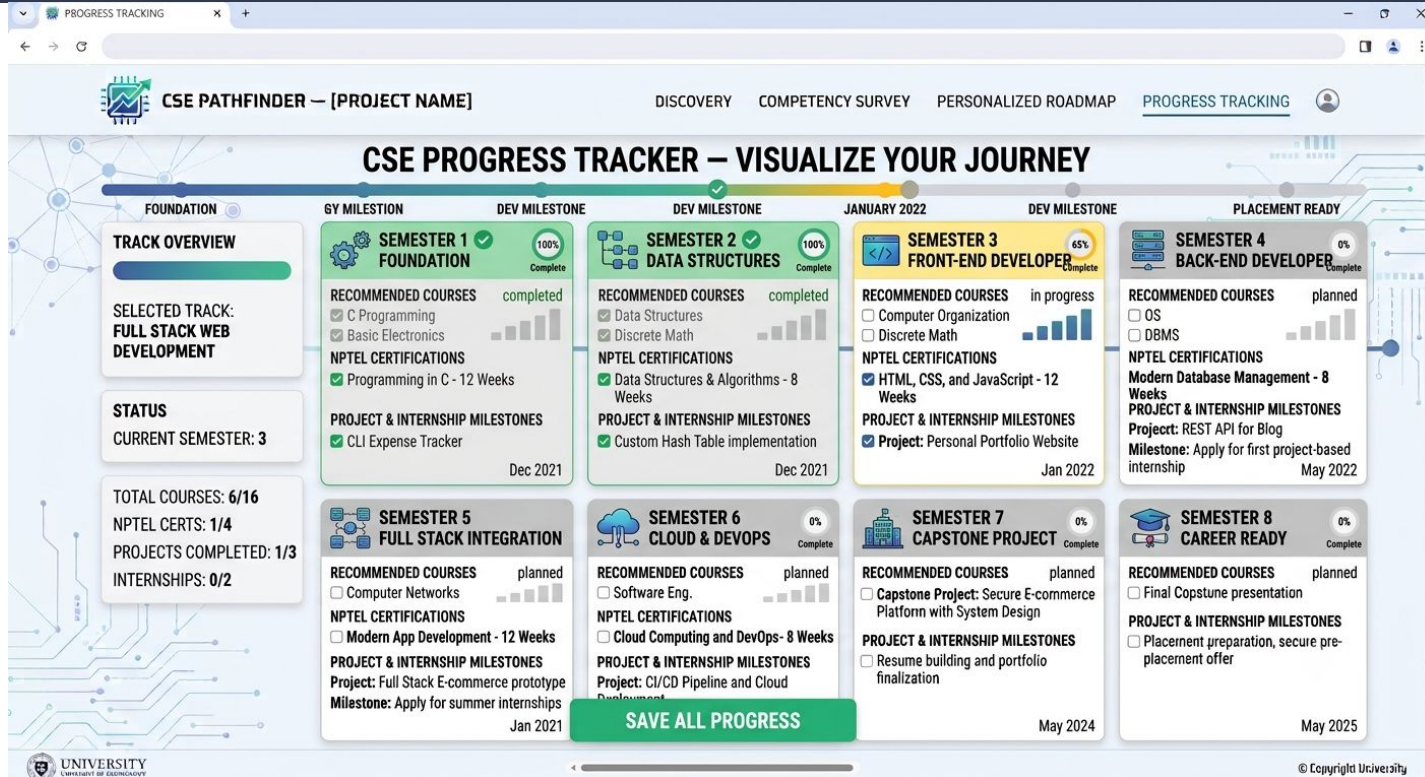
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GUI Design – Page 4



GUI Design – Page 5



Project Timeline

WEEK 1 — PLANNING & RESEARCH

Requirements • CSE Specializations • Industry Analysis

WEEK 2 — SYSTEM & UI DESIGN

Wireframes • Architecture • MongoDB Schema

WEEK 3–4 — BACKEND DEVELOPMENT

Spring Boot • REST APIs • Authentication • Database

WEEK 5 — SPECIALIZATION DISCOVERY

Specialization Details • Industry Demand • Career Paths

WEEK 6 — COMPETENCY SURVEY

Questionnaire • Skill Assessment • Competency Level

WEEK 7–8 — PERSONALIZED ROADMAP

Semester Plan • NPTEL • Projects • Internships

WEEK 9 — PROGRESS TRACKING

Milestones • Progress Dashboard • Achievements

WEEK 10 — TESTING & DEPLOYMENT

Integration Testing • Bug Fixing • Deployment • Documentation

FLOW:

PLAN → DESIGN → DEVELOP → DISCOVER →
ASSESS → PERSONALIZE → TRACK → DEPLOY

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